5. **Boron (B):** Translocation of carbohydrates, uptake and utilisation of Ca^{++}, pollen germination, nitrogen metabolism, fat metabolism, cell elongation and differentiation. It is absorbed as (borate) BO^{3-} ions.

   Deficiency: Death of root and shoot tips, premature fall of flowers and fruits, brown heart of beet root, internal cork of apple and fruit cracks.

6. **Molybdenum (Mo):** Component of nitrogenase, nitrate reductase, involved in nitrogen metabolism, and nitrogen fixation. It is absorbed as molybdate (Mo^{2+}) ions.

   Deficiency: Chlorosis, necrosis, delayed flowering, retarded growth and whip tail disease of cauliflower.

7. **Chlorine (Cl):** It is involved in Anion – Cation balance, cell division, photolysis of water. It is absorbed as Cl^{-} ions.

   Deficiency: Wilting of leaf tips

8. **Nickel (Ni):** Cofactor for enzyme urease and hydrogenase.

   Deficiency: Necrosis of leaf tips.



Calmodulin

Calmodulin is a Ca^{2+} modulating protein in eukaryotic cells. It is a heat stable protein involved in fine metabolic regulations.

Activity

Collect leaves showing mineral deficiency. Tabulate the symptoms like Marginal Chlorosis, Interveinal Chlorosis, Necrotic leaves, Anthocyanin formation in leaf, Little leaf and Hooked leaf. (Discuss with your teacher about the deficiency of minerals)

12.4 Deficiency diseases and symptoms

The following table (Table 12.2) gives you an idea about Minerals and their Deficiency symptoms:

Table 12.2: Deficiency diseases and Symptoms

Name of the deficiency disease and symptoms	Deficiency minerals
1. Chlorosis (Overall)	Nitrogen, Potassium, Magnesium, Sulphur, Iron, Manganese, Zinc and Molybdenum.
a. Interveinal chlorosis	Magnesium, Iron, Manganese and Zinc
b. Marginal chlorosis	Potassium
2. Necrosis (Death of the tissue)	Magnesium, Potassium, Calcium, Zinc, Molybdenum and Copper.
3. Stunted growth	Nitrogen, Phosphorus, Calcium, Potassium and Sulphur.
4. Anthocyanin formation	Nitrogen, Phosphorus, Magnesium and Sulphur
5. Delayed flowering	Nitrogen, Sulphur and Molybdenum
6. Die back of shoot, Reclamation disease, Exanthema in citrus (gums on bark)	Copper
7. Hooked leaf tip	Calcium
8. Little Leaf	Zinc
9. Brown heart of Beet root and Internal cork of apple	Boron
10. Whiptail of cauliflower and cabbage	Molybdenum
11. Curled leaf margin	Potassium

XIth_Botany_Chapter_12_EM.indd 263 5/17/2020 9:19:37 PM